Math 1610, Honors Calculus 1, Fall 2026, Homework 7

1. (Limits of vector-valued functions.) Let $\vec{F}, \vec{G}$ be vector-valued functions with values in $\mathbb{R}^n$, defined on an open interval containing c (except possibly at c itself), and suppose

$$\lim_{t \to c} \vec{F}(t) = \vec{L}, \qquad \lim_{t \to c} \vec{G}(t) = \vec{M}.$$

   (a) Show that $\big|\, \|\vec{x}\| - \|\vec{y}\| \,\big| \leq \|\vec{x} - \vec{y}\|$ for all $\vec{x}, \vec{y} \in \mathbb{R}^n$, and deduce that $\lim_{t \to c} \|\vec{F}(t)\| = \|\vec{L}\|$.

(b) Show, directly from the definition of the limit and without working in coordinates, that $\lim_{t\to c} \vec{F}(t) \cdot \vec{G}(t) = \vec{L} \cdot \vec{M}$.

(Write $\vec{F} \cdot \vec{G} - \vec{L} \cdot \vec{M} = (\vec{F} - \vec{L}) \cdot \vec{G} + \vec{L} \cdot (\vec{G} - \vec{M})$ and use Cauchy–Schwarz.)

2. (Computing gradients.)

   (a) Assuming that the gradients exist, use partial derivatives to compute ∇f, ∇g and ∇h, where

$$f(x,y,z) = xy + yz + zx, \qquad g(x,y) = \frac{x^2}{1+y^2}, \qquad h(\vec{x}) = \|\vec{x}\|^4 \quad (\vec{x} \in \mathbb{R}^n).$$

(b) Define $k : \mathbb{R}^2 \to \mathbb{R}$ by

$$k(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0). \end{cases}$$

Compute both partial derivatives of k at the origin. Then show that k is not differentiable at the origin.

(The partials tell you the only possible value of $\nabla k(\vec{0})$. Test that value in the definition, letting $\vec{h}$ approach $\vec{0}$ along the line $y = x$.)

3. (Limits in the plane.)

   (a) Show, directly from the definition of the limit, that

$$\lim_{(x,y)\to(0,0)} \frac{x^3 - y^3}{x^2 + y^2} = 0.$$

(b) Define $f : \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0). \end{cases}$$

Show that both partial derivatives of f exist at the origin, but that f is not continuous there.

4. (The length function.) Let $N : \mathbb{R}^n \to \mathbb{R}$ be $N(\vec{x}) = \|\vec{x}\|$.

   (a) Show, directly from the definition, that N is continuous at every point of $\mathbb{R}^n$.

(b) Let $\vec{a} \neq \vec{0}$. Show, directly from the definition of the gradient, that N is differentiable at $\vec{a}$, with $\nabla N(\vec{a}) = \vec{a}/\|\vec{a}\|$.

(Multiply and divide $\|\vec{a}+\vec{h}\| - \|\vec{a}\|$ *by* $\|\vec{a}+\vec{h}\| + \|\vec{a}\|$*.)*

(c) Show that N is not differentiable at $\vec{0}$.